sons more or less fantastic for this supposed superiority of the wrought iron axle.

The matter was discussed in a committee report in 1894 to this effect:

"The opinions of roads with regard to the comparative freedom from heating of iron and steel axles, seem to be about equally divided, and about all that can be said is that steel axles are freer from flaws and seams in journals than iron, and for that reason heat less from those causes. But, on the other hand, the grain of steel axles being much closer than in iron, there is less opportunity for the oil to be held in the minute spaces between the molecules of the metal, and, as a consequence, it is somewhat more difficult to lubricate steel journals, if they are loaded anywhere to their full capacity."

The probability is that any apparent differences that may have appeared were, for the most part imaginary. However that may be, the steel axle has entirely supplanted the older iron axle and it is doubtful if it will ever come back to its own again.

The matter of axles coming before the association was in 1869 when there was a long discussion, followed by the adoption of a resolution that "an axle for heavy passenger and sleeping cars should be at least 4½ in. in the wheel seat, and 3¾ in. in the center, the journals to be 3¼ in. by 7 in. long and of the best hammered iron."

This was an expression of opinion and not the formal adoption of a standard. Then the matter was dropped for two years to be brought up at the 1872 convention where a verbal report was made by a committee and there were tentative recommendations of a form of axle somewhat resembling the present shapes, but of much smaller dimensions. At first the suggestions were merely towards a uniformity of axles and oil boxes but before the meeting was over, they were talking about standards; the first man recorded as having used the term being Mr. M. C. Andrews of the New York & New Haven Railroad, and the size of journals recommended was 3¼ in. in diameter and 5½ in. or 6 in. long. This discussion closed with the appointment of a committee to report on car axles, oil boxes, journal bearings and center plates at the next meeting. This was in 1872.

The next year the committee made a report and recommended for adoption as a standard an axle 4½ in. in diameter at the wheel seat, and with a journal 3½ in. in diameter and 7 in. long. Mr. Adams of the committee presented a minority report suggesting a larger diameter of journal but giving no measurement. The matter was considered of such importance that discusion was postponed for 24 hours, when Mr. Adams came out with a definite recommendation to adopt 4 in. for the journal diameter. Then the discussion was on. The representative of the Pennsylvania argued for a smaller journal; he knew that 3½ in. was big enough because it and even smaller axles were running satisfactorily, and that such a big journal as 4 in. would not be used. Mr. Leander Gary of the New York Central was a prominent member in those days and his argument was the essence of good sense. He said: "I hope the gentlemen will not adopt anything because they think the Pennsylvania road or the Erie road or the New York Central road or any other railroad will adopt it, but will recommend something that will be right, if it is never adopted." The discussion occupied nearly the whole of the second day's proceedings and was continued on the third. For hours the discussion waged, the advocates for a small journal led by Mr. Van Houten of the Pennsylvania and those for the large by Mr. Gary of the New York Central. Finally after ballots for choice of dimensions one was taken on the recommendation of the committee for a 3½ in. by 7 in. journal. This was lost by a vote of 29 to 38. Then came the final vote on the 3¾ in. by 7 in. recommendation which was adopted unanimously, and the first standard of the Master Car Builders' Association was born.

It is interesting to note that in this whole discussion there was not one word about fiber stresses on the axle. It is doubtful if there was a man there that ever thought of it or of elastic limits and such like things. It was a discussion of purely practical men who were guessing at results and who fortunately guessed well. But it was a long time before the standard came to be accepted. In 1876 it was suggested to rescind the vote of 1873, and as late as 1878, there were a great variety of axles in use on many of the roads of the country; the Canada Southern alone reporting the possession of 18 classes.

In all of this there was nothing about specifications and tests of materials, nothing about stresses and the action under load. It was merely a matter of getting an axle big enough not to break, and one that would run cool in the journals. Hot boxes had been the bug-a-boo of the advocates of the small boxes and it took much argument to convince them that the higher peripheral speed of the larger journal could be made to run cool.

It must be borne in mind that at the time of the adoption of the first standard axle, the capacity of freight cars throughout the country was ten tons, without an allowable percentage of overload; that the weight of the empty car was the same and that there was no intention or expectation of increasing that capacity. The standard was adopted by the individual vote of the members present regardless of the number of cars under their jurisdiction, so that it is small wonder that there was a minority, opposing the increase in the general practice as to the size of axles when the discussion was filled with records of axles worn to all diameters down to 2 in. that had done their work without failure.

There were no tests and no specifications for axles proposed for purchase or inspection, and it was tacitly assumed that they would be of wrought iron. Indeed it was not until 1896 that the matter came up.

Meanwhile in 1886 a specification was drawn up by Mr. Theo. N. Ely, superintendent of motive power of the Pennsylvania R. R. for iron and steel axles which latter had, as we have seen, then become the standard axle material of the road. And these specifications may be regarded as the root from which later specifications were developed. They were as follows:

Specifications for Axles.

For each 100 axles ordered, 101 must be furnished, from which one will be taken at random, and subjected to tests prescribed for such axles. If the axle stands the prescribed test, the 100 axles will be carefully inspected and those only will be accepted which are made and finished in workmanlike manner and which are free from cracks or unwelded seams.

Locomotive tender and car axles to have journal swaged, and all axles to be centered with 60 degree centres.

Passenger Car and Passenger Locomotive and Tender Truck Axles.

Axles must be made of steel and be rough turned throughout.

Two test pieces will be cut from an axle, and the test sections of ⅝ inch diameter by 2 in. long may fall at any part of the axle, provided, that the centre line of the test section is one inch from the centre line of the axle. Such test pieces should have a tensile strength of 80,000 lbs. per square inch and an elongation of 20 per cent. Axles will not be accepted if the tensile strength is less than 75,000 lbs., nor the elongation below 15 per cent., nor if the fractures are irregular.

Freight Car and Freight Locomotive and Tender Truck Axles.

Steel.—Steel axles for freight cars, and freight locomotive, and tender trucks, will be subjected to the following test, which they must stand without fracture:

Axles 4 inches diameter at centre—Five (5) blows at 20 feet of a 1640 lbs. weight, striking midway between supports 3 feet apart; axle to be turned over after each blow.

Axles 4⅜ inches diameter at centre—Five (5) blows at 25 feet of a 1640 lbs. weight, striking midway between supports

662. Q.—At what other time should emergency position be used?

A.—In case of a parted passenger train, or brakes on a passenger train applying in emergency.

663. Q.—What does the equalizing discharge valve do?

A.—Mechanically measures the volume of air discharged from the brake pipe during service operations.

664. Q.—What three principal pressures are divided by the brake valve on lap position?

A.—Main reservoir, brake pipe and equalizing reservoir.

665. Q.—What part of the brake valve separates the pressures?

A.—The rotary valve and seat and the equalizing piston packing ring.

666. Q.—What pressures does the packing ring divide?

A.—Brake pipe and equalizing reservoir.

667. Q.—What then occurs as the brake valve is moved to service position?

A.—The equalizing reservoir pressure is reduced through the preliminary exhause port, and brake pipe pressure lifts the equalizing piston and flows to the atmosphere.

668. Q.—How long does the piston remain off its seat?

A.—Until the brake pipe pressure has reduced to a point slightly less than that remaining in the equalizing reservoir when the valve handle was returned to lap position.

669. Q.—What are the movable parts of the application portion of the distributing valve?

A.—An application piston, and application valve and an exhaust valve.

670. Q.—What positions do they assume in operation?

A.—Release, lap and application position.

671. Q.—What are the movable parts of the equalizing portion?

A.—The equalizing piston, equalizing slide valve and graduating valve, the emergency slide valve and brake cylinder check valve, and a piston valve in the safety valve.

672. Q.—What positions does it assume?

A.—Release, service, service lap, emergency and emergency lap.

673. Q.—What pressure is the safety valve of the distributing valve adjusted for in passenger service?

A.—68 lbs.

674. Q.—In freight service?

A.—The same.

675. Q.—What pressure is exhausted to the atmosphere by the safety valve?

A.—Application cylinder and application chamber pressures.

676. Q.—How does this control the pressure developed and retained in the brake cylinders?

A.—The pressure in the application cylinder governs the amount that will be developed in the brake cylinders.

677. Q.—What air pressures are separated by the application valve of a distributing valve?

A.—Main reservoir and brake cylinder.

678. Q.—By the application piston packing leather and ring?

A.—Application cylinder and brake cylinder.

679. Q.—By the equalizing piston and packing ring?

A.—Brake pipe and pressure chamber pressure.

680. Q.—By the exhaust valve, with the brakes applied?

A.—Brake cylinder and atmospheric.

681. Q.—What is the purpose of the application chamber of a distributing valve?

A.—An enlargement of the application cylinder volume for service operation.

682. Q.—What principal pressures are separated by the equalizing slide valve with the brake applied?

A.—Pressure chamber and application chamber pressures.

683. Q.—If the equalizing portion is in quick action position?

A.—Application cylinder and application chambers are also separated.

684. Q.—What pressures are separated by the emergency slide valve of the quick action equalizing cylinder cap?

A.—Brake pipe and brake cylinder pressures.

685. Q.—By the cylinder check valve, with the brake applied?

A.—Brake pipe and brake cylinder pressures.

686. Q.—How is a distributing valve cut out?

A.—By closing the stop cock in the distributing valve supply pipe.

687. Q.—How is it positively cut out, if it has a quick action cylinder cap?

A.—By closing the stop cocks in the brake cylinder pipes.

688. Q.—Why cannot the latter valve be positively cut out with the stop cock in the supply pipe alone?

A.—Because with an emergency application, the emergency slide valve of the quick action cylinder cap will be moved and admit brake pipe pressure to the brake cylinders.

689. Q.—Which is the larger of the two chambers of the distributing valve reservoir?

A.—The pressure chamber.

690. Q.—How does a brake pipe reduction affect a distributing valve?

A.—If sufficiently rapid, it reduces brake pipe pressure below that in the pressure chamber, permitting the higher pressure to move the equalizing portion to application position.

691. Q.—How does the brake then apply?

A.—Pressure chamber pressure flowing through the service port to the application chamber and application cylinder, forces the application piston to application position admitting pressure from the main reservoir to the brake cylinders.

692. Q.—How long will main reservoir pressure continue to flow into the brake cylinders?

A.—Until brake cylinder and application cylinder pressures become equal.

693. Q.—What then occurs?

A.—The application piston graduating spring moves the application portion to lap position.

694. Q.—How long will the portion remain in lap position?

A.—Until leakage reduces brake cylinder pressure sufficiently to permit the application cylinder pressure to again move the valve to application position, thus supplying the leakage.

695. Q.—How is the brake then released?

A.—By exhausting application cylinder pressure to the atmosphere.

696. Q.—In what positions must the brake valves be to permit application cylinder pressure to exhaust?

A.—Both handles must be in running position, otherwise the independent valve must be moved to release position.

697. Q.—What is the principle upon which an automatic brake operates?

A.—Upon the creation of differences in pressure between stored volumes of compressed air.

698. Q.—What are the limits for brake cylinder piston travel?

A.—4 to 6 ins. for the drivers and 6 to 8 ins. on the tender.

699. Q.—Will a brake release quicker when the travel is long or short?

A.—The locomotive and tender brakes will release quicker if the travel is short.

700. Q.—Why so?

A.—There will be less volume in the cylinder to exhaust than when the travel is long.

701. Q.—In service application, will brake cylinder pressure on cars and engine be uniform?

A.—Yes, if brake cylinder piston travel on cars is approximately 7½ or 8 ins.

702. Q.—Can the automatic brake valve be used to make a service application if the independent brake valve is in release position?

A.—No. Application cylinder pressure would escape through the independent brake valve exhaust port.

703. Q.—What returns the brake cylinder pistons to release position after the exhaust of air pressure from the cylinders?

A.—The cylinder release springs.

704. Q.—What holds the brake cylinder packing leathers against the cylinder wall?

A.—The packing expander ring.

(To be continued.)

alone, how many branches has the reservoir pipe?

A.—Usually five.

836. Q.—To what parts do they supply main reservoir pressure?

A.—To the brake valve direct, to the feed valve, reducing valve, air gage, pump governor, distributing valve and dead engine fixture if one is used.

837. Q.—What reservoir pipe connections are sometimes omitted?

A.—The one to the feed alve when the feed valve is attached to the H6 brake valve, the one to the dead engine fixture when this part is omitted, and the governor connection is sometimes made to the reservoir connection pipe or to the main reservoir direct.

838. Q.—How is the engine hauled dead in a train if the dead engine fixture is omitted?

A.—By leaving the brake valve cut out cock open, unscrewing the adjusting nut of the brake pipe feed valve to maintain about 10 lbs. pressure and plugging the brake valve service exhaust port.

839. Q.—Should anything be done to limit the brake cylinder pressure?

A.—The adjusting nut of the safety valve of the distributing valve should be slacked off to prevent more than 30 or 35 lbs. pressure in the applicaiton cylinder and consequently the brake cylinders.

840. Q.—What is the object?

A.—To prevent the possibility of wheel sliding when no one is riding the engine.

841. Q.—With the equipment of the dead engine arranged as stated, how is the main reservoir charged?

A.—Through the open brake valve cut out cock, and through the brake pipe feed valve in the reverse direction.

(To be continued.)

Train Handling.

(Continued from page 180, June, 1919.)

855. Q.—What should be done with hose couplings after they have been used in this manner?

A.—They should have the couplings gauged before being returned to service.

856. Q.—What can be done if the brake pipe is bursted or broken under the tender?

A.—The brake and signal hose may be connected front and rear of tender and the signal pipe used to convey brake pipe pressure from the engine to the brake pipe of the cars.

857. Q.—Can the same thing be done in the event of a broken brake pipe on a passenger car?

A.—Yes, but the car discharge valve should be closed.

858 Q.—For what purpose?

A.—To prevent the possibility of anyone attempting to use the signal whistle from this car and make a brake pipe opening instead.

859. Q.—How would this affect the signal whistle?

A.—It would be operative on the train ahead of the broken brake pipe, but inoperative back of it.

860. Q.—Sometimes the brake pipe branch to the triple valve is broken on a car near the triple valve; what should be done?

A.—Close the stop cock in the branch pipe and bleed the auxiliary reservoir or other air reservoirs that may be on the car.

861. Q.—Sometimes the brake pipe branch is broken off at the stop cock but on the main brake pipe side.

A.—In this case it may be possible to remove the broken piece og pipe and unscrew the cut out cock and good nipple and screw this into the main brake pipe, closing the stop cock.

862. Q.—What would be done if the supply pipe to the distributing valve was to break off?

A.—If near the distributing valve reservoir, the supply pipe cut out cock would be closed, if between the cut out cock and main reservoir pipe it would be necessary to plug the pipe to stop the main reservoir leak.

863. Q.—Would there be a driver brake under this condition?

A.—No; engine and tender brakes would be inoperative.

864. Q.—What might be done if time permitted and both ends of the broken pipe could be removed?

A.—It might be possible to make a connection with two signal hose.

865. Q.—Where could they be secured if there was none in the tank box?

A.—One from the pilot and the other from the rear end of the train.

866. Q.—Is the thread on the signal hose same as the threads in the fittings of the supply pipe?

A.—Yes, both are standard ¾-in. pipe thread.

867. Q.—Would it be possible to operate the driver brake on a lone engine if the distributing valve supply pipe was broken and no connection could be made?

A.—Yes, but both ends of the broken supply pipe must be closed.

868. Q.—How would it be done?

A.—By unscrewing the cap nuts of the quick action cylinder cap and removing the emergency slide valve and cylinder check valve and replacing the cap nuts.

869. Q.—How would the brake valve handles be carried?

A.—The automatic in service application position and the independent valve in slow applicaiton position.

870. Q.—What would be the condition of the brake pipe and application portion of the distributing valve?

A.—The brake pipe would be empty and the application portion in applicaiton position.

871. Q.—How would the brake be applied?

A.—By moving the automatic brake valve to release position.

872. Q.—And the main reservoir pressure would flow?

A.—Into the brake pipe and through the quick action portion of the distributing valve into the brake cylinders.

873. Q.—How would the brake be released?

A.—With the independent brake valve after rturning the automatic brake valve to service position.

874. Q.—And after which?

A.—The independent valve is again placed in slow application position.

875. Q.—If more than 45 lbs. pressure was admitted to the brake cylinders would there not be a waste from the distributing valve exhaust port?

A.—Yes, but if more brake cylinder pressure was required, the reducing valve could be adjusted to the amount desirable to use.

876. Q.—What if the engine was equipped with a plain cylinder cap on its distributing valve, and the supply pipe was broken off, no repairs or connections could be made and it was necessary to have a driver and tender brake for the lone engine?

A.—The equalizing cylinder cap gasket could be removed and a passageway cut betewen the brake pipe and brake cylinder port. When replaced in the proper position, the brake would be operated as previously explained.

877. Q.—Could the brake on a train be operated while such a temporary brake was arranged for the engine?

A.—No.

878. Q.—What would be done if the brake cylinder pipe was to break off at the distributing valve?

A.—There would be no brake on the engine or tender, therefore the stop cock in the distributing valve supply pipe would also be closed.

879. Q.—For what purpose?

A.—To prevent the escape of main reservoir pressure when the brake was applied on the cars of the train.

880. Q.—What if the pipe broke off while running along the road and was first known of by a loss of main reservoir pressure while making a slight reduction?

A.—The independent brake valve could be placed in release position to prevent the loss of main reservoir pressure until the supply pipe cut out cock could be closed.

881. Q.—How would this prevent the loss of main reservoir pressure?

A.—Exhausting the pressure from the application chamber would permit the application piston graduating spring and the in-rush of main reservoir pressure to move the application piston to lap position.

882. Q.—What would be done if the brake cylinder pipe under the tender was broken?

A.—The stop cock in the pipe leading

636. Q.—What sizes of brake cylinders are they used with?

A.—The K-1 for 8 in. cylinders and the K-2 for 10 in.

637. Q.—Have these valves all of the features of the H valves?

A.—Yes, and certain other improvements in addition.

638. Q.—What features are added?

A.—Quick service, uniform recharge of the auxiliary reservoirs and a restricted release.

639. Q.—What is gained by the quick service?

A.—Each triple valve as it moves to service application position makes a local brake pipe reduction by admitting air from the brake pipe to the brake cylinder and this positively continues the reduction throughout the train.

640. Q.—With H valves on a 100 car train how many of the brakes can be applied with a 5 lb. brake pipe reduction?

A.—About 35 or 40 on the head end.

641. Q.—With a 10 lb. brake pipe reduction?

A.—Possibly 75 brakes.

642. Q.—Will a 15 lb. brake pipe reduction apply all of the operative brakes in a 100 car train?

A.—Not as a general proposition, a 15 lb. reduction will be required, but some of the brakes near the middle of the train will usually fail to apply?

643. Q.—Why so?

A.—Because all of the compressed air leaving the brake pipe, except that lost through leakage must pass through the brake pipe exhaust part of the brake valve and as the brake pipe pressure is being reduced at the head end of the train, the flow forward from the rear end tends to maintain the pressure in the middle of the train and the brake pipe reduction is so slow that the auxiliary reservoir pressure has a chance to reduce by flowing into the brake pipe through the triple valve feed grooves without moving the triple valve pistons.

644. Q.—In a practical way, what does the pipe friction or the frictional resistance to the flow of air amount to in brake operation?

A.—It delays and tends to destroy the application of brakes and tends to cause failures of the brakes to release.

645. Q.—To what extent does it retard the flow of air from the brake pipe?

A.—To such an extent that if the triple valves are put out so that they will not assist in the brake pipe reduction, the brake pipe pressure can be exhausted from a long brake pipe with the brake valve in service position as quickly as it can with the brake valve in emergency position.

646. Q.—Or in other words?

A.—There is enough resistance to the flow of air so that the volume of the brake pipe can be exhausted as quickly with a 9-32 in. circular opening as it can with a 5/8 in. opening.

647. Q.—Why so?

A.—Because it requires just so much time to expand this compressed air and withdraw it through the various cracks and turns in the piping.

648. Q.—What effect on the release of brakes or the admission to the brake pipe?

A.—That if an air gage is attached to the brake pipe 30 cars back in the train, the increase in brake pipe pressure will be so slow that from an observation of the gage hand it will be impossible to determine whether the brake valve is in release or in running position.

649. Q.—What is then the one insurmountable obstacle in the prevention of uniform brake operation on freight trains?

A.—The large volume of compressed air in the brake pipe.

650. Q.—What is the difference in the amount of air discharged from the brake valve during an application where K valves are used?

A.—Only about one-half of the volume is discharged as when H valves are used on the same length of train.

651. Q.—What becomes of the other half of the volume that would otherwise be discharged with H valves?

A.—It is admitted to the brake cylinders.

652. Q.—Does this make any material difference in the final brake cylinder pressure obtained?

A.—No, only two or three pounds more will be obtained.

653. Q.—Will this reduction added to that of the brake valve not tend to produce undesired quick action?

A.—No, when the rate of brake pipe discharge is almost equal to the capacity of the triple valve service port to expand auxiliary reservoir pressure, the triple valve will be moved to full service position and the quick service part will not be operative.

654. Q.—How does the valve then operate on a short train?

A.—The valve does not assume quick service position but momentarily, or when the quick service feature is not required to hasten the brake pipe reduction. It is automatically cut out.

655. Q.—What is meant by retarded release?

A.—When the brake pipe pressure is three or more pounds in excess of or higher than the pressure in the auxiliary reservoir, the retarding spring is compressed and the triple valve is moved to restricted release position, in which the exhaust of brake cylinder pressure is restricted.

656. Q.—To what extent?

A.—So that the brakes on the rear of a moderate length of train will release uniformly with the brakes on the head end.

657. Q.—How far back in the train can this difference in pressure be obtained to force the triple valves to this position?

A.—As far back as 25 or 30 cars if the brake valve is allowed to remain in release position for about 15 seconds.

658. Q.—What does this restricted release do?

A.—Prevents the slack from running out when a release of brakes is made or assists the engine brake to hold in the slack during a release.

659. Q.—How is uniform recharge secured?

A.—When the triple valve is in retarded release position one of the feed grooves is also closed and the auxiliary reservoirs can charge but about one-half as fast as when the valve is in normal release position with both grooves open.

660. Q.—What returns the triple valves at the head end to normal release position?

A.—When the brake pipe and auxiliary reservoir pressure are within 3 lbs. of each other, the springs of the retarding device return the triple valve piston to normal release position.

661. Q.—How much of a brake pipe reduction will be required to apply the brakes on a 100 car train with these types of triple valves?

A.—A five pound reduction will be propagated through the entire train and will apply all of the operative brakes.

662. Q.—When a car weighing say 40,000 lbs. is equipped with a brake what will be the percentage of braking ratio?

A.—Usually 60 per cent. of the light weight of the car which is then based upon a 50 lb. pressure in the brake cylinder.

663. Q.—Why is the ratio so much lower than for passenger cars?

A.—The speed of freight cars is much slower and the slower the speed, everything else being equal, the more effective the brake shoe is in retarding the speed of the wheel.

664. Q.—Why is this?

A.—Because a brake shoe when closely examined will be found to have abrasions or elevations and depressions all over the bearing surface and at slow speeds these uneven surfaces tend to interlock while at high speeds the rapid turning of the wheels tend to break them off and the shoe has much less frictional effect.

665. Q.—If the car is braked as outlined and the car is then loaded, say with about 100,000 lbs., what will the percentage of braking ratio be, or that which is generally termed percentage of braking power?

A.—About 17 per cent. of the total weight of load and car.

(To be continued.)

out cock that opens the reservoir pipe to the atmosphere when the cock handle is moved to the closed position.

952. Q.—How does placing the brake valve in release position prevent the unseating of the rotary valve?

A.—By making a direct opening from the brake pipe to the reservoir pipe.

953. Q.—How would the feed valve be affected through a failure to move the brake valve to release position before closing the reservoir cut out cock?

A.—The supply valve might be unseated through the reservoir pressure suddenly leaving the supply valve bushing.

954. Q.—What would likely result from these valves becoming unseated under the conditions mentioned?

A.—Dirt or foreign matter lodging on the valve seats might start them to leaking.

955. Q.—How is this matter handled when the brake valve cut out, or double heading cock is located in the reservoir pipe?

A.—The cock is not turned, but the brake valve handle is placed in release position before the main reservoir cut out cock is closed.

956. Q.—How many pipe connections has the brake valve cut out cock, when located in the reservoir pipe?

A.—Three. The reservoir pipe, reservoir gage pipe, and the brake valve service exhaust pipe.

957. Q.—What additional pipe did the previous types of cocks have?

A.—A pipe leading from the brake pipe to the large end of the cock, usually called balance pipe.

958. Q.—What was it used for?

A.—To assist the spring in the cock in holding the plug valve to its seat.

(*To be continued.*)

Train Handling.

(*Continued from page 278, Sept.,* 1919.)

975. Q.—Are broken air pipes with the LT equipment handled the same as with the ET equipment?

A.—Practically in the same manner but with a few slight exceptions.

976. Q.—What is one of them?

A.—With a broken control valve supply pipe, the locomotive and tender brake can still be applied with the straight air brake.

977. Q.—And while the train brakes are being applied with the automatic valve?

A.—The straight air can be used on the engine and tender.

978. Q.—How should the leak from the broken supply valve pipe be stopped?

A.—By closing the cut out cock at one end and plugging the other.

979. Q.—Why close both ends of the broken pipe?

A.—To prevent a waste of air if the brake is applied in emergency or with a heavy or over reduction in brake pipe pressure.

980. Q.—Where would this waste be from?

A.—Out of the brake pipe, through the quick action cylinder cap and through the broken supply pipe.

981. Q.—If a distributing valve supply pipe was then broken off would it not be of advantage to plug both ends of the broken pipe, that is, closing the stop cock in one end and plugging the other?

A.—Yes, it would provide a brake for the engine and tender for emergency applications of the brake, also one in case of an over reduction in brake pipe pressure.

982. Q.—In what manner?

A.—By a flow of air from the brake pipe through the quick action equalizing cylinder cap into the brake cylinders when the equalizing valve moves to emergency position.

983. Q.—Are brake valve equalizing reservoir pipe, gage pipes and governor pipes if broken with the LT handled same as with the ET?

A.—Yes.

984. Q.—And with a broken off brake pipe forward or back of the brake valve branch on passenger engines?

A.—Yes, they are handled in the same manner.

985. Q.—What if the brake pipe is broken off at the automatic brake valve, or if broken back of the brake valve branch on a freight engine?

A.—The straight air brake valve will be used to operate the train brakes.

986. Q.—How will the connection be made?

A.—Between the brake cylinder hose rear of engine and the brake pipe hose front of tender.

987. Q.—What would be the first thing to be done when the broken off pipe at the brake valve exhausted the brake pipe pressure and stopped the train?

A.—The automatic brake valve handle would be placed on lap position to hold the main reservoir pressure and the brake on the engine would be released with the straight air valve in automatic release position.

988. Q.—And thereafter?

A.—The stop cocks in the brake cylinder pipe leading to the driver brake, truck brake and other cylinders, except the one toward the tender cylinder would be closed.

989. Q.—And then?

A.—The hose couplings mentioned would be hammered together and the straight air brake valve placed in full application position, the safety valve adjusting nut screwed down and the pressure increased by screwing in on the adjusting nut of the reducing valve.

990. How should this be done?

A.—The pressure for the straight air pipe used as a brake pipe should be adjusted with the angle cock at the rear of the tender closed and the cock opened afterwards.

991. Q.—And what should be done while the brakes are releasing and the brake system is charging for a brake test?

A.—The stop cock in the control valve supply pipe should be closed, and the union connection on the automatic side of the double check valve should be slacked off in order to permit a leak if air accumulated in the pipe at the automatic side of the check valve.

992. Q.—The object of closing the supply pipe cock?

A.—To prevent the possibility of main reservoir pressure coming through the application portion of the control valve into the brake cylinder pipe.

993. Q.—And of slacking off the union connection?

A.—To keep this pipe free of air pressure.

994. Q.—What might be the result of pressure bottling up in the brake cylinder pipe?

A.—The double check valve would be shifted and cut the straight air brake valve off from the brake pipe.

995. Q.—How would the train brakes be applied, after these connections have been made?

A.—By moving the straight air brake valve to release position.

996. Q.—Exhausting pressure from where?

A.—The brake cylinder pipe of the engine and the brake pipe of the train.

997. Q.—And if the check valve should flip over from pressure built up in the brake cylinder side of the check valve?

A.—It would prevent the application of the brake.

998. Q.—And the normal position of the straight air valve while running along the road?

A.—Full application position, admitting the maximum amount of air from the main reservoir to the brake pipe.

999. Q.—What controls brake pipe pressure?

A.—The reucing valve.

1000. Q.—Can this same arrangement be used if both the reservoir pipe and feed valve pipe were to be broken off at the brake valve at the same time?

A.—Yes, after stopping the main reservoir leak.

1001. Q.—On a lone engine, why will the brake generally apply if the brake valve handle is moved to release position and left there for some little time?

A.—Because the exhaust port of the equalizing slide valve is closed and any